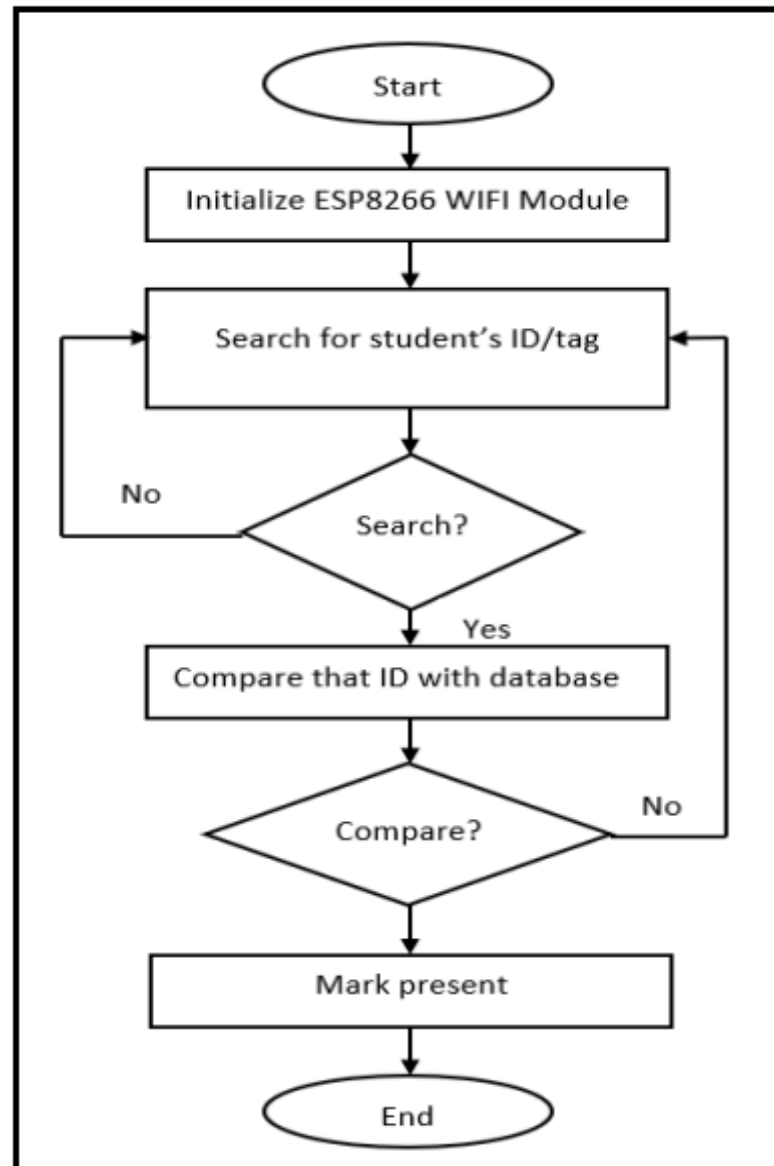


4. Flowchart

[Flowchart for NFC/RFID Attendance System]

1. User scans NFC/RFID tag
2. Tag ID is verified
3. Attendance is logged
4. Data is updated in Google Sheets in real-time
5. System ready for next scan



5. Problem Statement

Ram needs a cost-effective and simple attendance system for his school that provides easy access to attendance data. He cannot afford a separate, complex database management

7. List of Components (Specifications)

NodeMCU ESP8266

- **Purpose:** Wi-Fi enabled microcontroller for connecting to the internet and sending data to Google Sheets.
- **Specification:**
 - Voltage: 3.3V
 - Wi-Fi: 802.11 b/g/n
 - I/O Pins: 9 digital pins, 1 analog pin
 - Operating Frequency: 80MHz

PN532 NFC Module

- **Purpose:** NFC/RFID reader for scanning tags.
- **Specification:**
 - Interface: I2C
 - Operating Voltage: 3.3V - 5V
 - Range: 0-5 cm
 - Frequency: 13.56 MHz

Google Spreadsheet Integration

- **Purpose:** Real-time data storage and tracking of attendance.
- **Specification:**
 - Cloud-based storage system
 - Accessible from any device with internet access
 - Real-time data update through Google Scripts

Buzzer (3V - 6V)

- **Purpose:** Provides audible confirmation for successful tag scans.
- **Specification:**
 - Operating Voltage: 3V - 6V
 - Current: <20mA

8. Technical Features

1. **Wi-Fi Connectivity:** Uses NodeMCU for seamless connection to the internet.
 2. **Real-Time Data Logging:** Attendance data is updated instantly in Google Sheets.
 3. **NFC/RFID Technology:** Allows contactless attendance marking for improved user experience.
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9. Project Features

- 1. **Cost-Effective:** Uses inexpensive NFC/RFID tags and open-source tools.
 - 2. **Real-Time Access:** Attendance data can be viewed from anywhere via Google Sheets.
 - 3. **Scalability:** Can be easily scaled for larger institutions by adding more NFC readers.
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5. Unit Testing

Feature	Before Unit Testing	After Unit Testing
Code Quality	High risk of bugs and errors in the code	Reduced bugs, high code reliability
Time for Debugging	Extensive time spent identifying and fixing bugs	Faster debugging process, minimal time spent
Code Coverage	Limited code coverage, potential untested scenarios	Comprehensive code coverage ensuring all functions are tested
Error Handling	Poor error handling with frequent unexpected crashes	Improved error handling with well-tested edge cases

10. Ways Forward

In the future, we plan to enhance the system by adding a **message alert system** that will notify users of their attendance status via SMS or email, providing an additional layer of convenience and accessibility.